Novel View Refinement via Attention Fusion

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Abstract

Novel view synthesis (NVS) aims to generate new views of an object given a single input view. While recent diffusion based methods like SV3D have shown impressive results, the task remains challenging, especially for views far from the input due to the limited information provided. We propose Novel View Refinement (NVR), a two-stage framework to improve the quality of synthesized novel views. First, we use a pretrained SV3D model to generate novel views from a front view of the object. Then, we input a new back view into the same frozen model, using the attention features from the first stage to guide the refinement process via fusion layer architecture. The fusion layers efficiently inject the knowledge from the first stage into the second stage while only requiring training a small number of additional parameters. Our experiments on the Objaverse dataset demostrate that NVR significantly improves the quality of the novel views generated by SV3D in a parameter efficient manner. We also open-source our implementation of the SV3D training code and Objaverse data processing pipeline to facilitate further research.

1. Introduction

Novel View Synthesis (NVS) is a task to generate unseen novel views from a single known view of an object. Recent works [7, 8] has shown outstanding performance for novel view synthesis utilizing diffusion-based approaches. Despite significant progress, the task is fundamentally considered an ill-posed problem because the models must infer unknown information from the limited context provided by the front view. Furthermore, views that are far from the given perspective face even greater challenges due to the shortage of information. Therefore, recent state-of-the-art NVS models often show unsatisfactory results compared to the ground truth, particularly for the views near the back side.

An intuitive approach to improve performance is to fine-tune the model to accept two or more images as inputs, providing more information about the object. However, fine-tuning the model in this manner is challenging as it alters all input dimensions and conditions. Additionally, it requires significant computational resources(i.e. VRAM and SSD), to train and store all the parameters of the baseline model.

Thus, we propose a simple, efficient, yet powerful framework called Novel View Refinement (NVR), which refines the synthesized novel views generated by the SV3D model by training only small add-on layers. Our approach involves two stages: First, we perform vanilla inference of SV3D using the front view and obtain the spatial attention features from the intermediate layers of the Unet. Second, we refine the results using both a new back view and the attention features obtained during the first stage. This approach is intuitive and expected to work well, as it provides nearly complete information—the front and back view—about the object to the model.

Therefore, we aim to realize the framework as an add-on framework, which can be completely separated from the baseline model, with the trained parameters being as lightweight as possible. This approach is motivated by preceding works such as [3, 9]. Specifically, we train lightweight fusion layers that fuse the attention features from the two stages, injecting the information from the front view into the back-view inference process.

Our framework is highly efficient since it trains only the additional fusion layers which have only $1/10000$ (150K) of the parameters of the baseline models (1.5B) which remain frozen during the training. Furthermore, we demonstrate the effectiveness of our approach in refining not only the front and back views but also the side views, as the front and back views implicitly provide the overall structure of the object.

Our contributions are summarized as follows:

- We propose a simple yet effective framework, Novel View Refinement (NVR), which refines the novel views generated by SV3D using an additional view.
- We propose an efficient add-on architecture, fusion layers, which leverage the knowledge of the baseline model for NVR with 1/10000 parameters compared to the baseline.
- Our experiments demonstrate that our framework effec-

*indicates equal contribution.

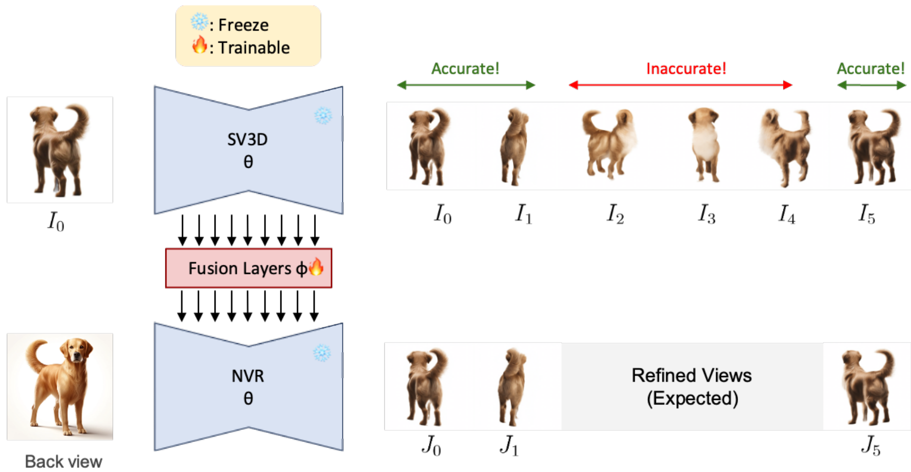

Figure 1. Conceptual illustration of Novel View Refinement. NVR model has frozen, same parameters as the original SV3D, while trainable Fusion Layers inject the attention maps of SV3D into NVR. NVR model refines the novel views using a new back view.

tively refines the novel views synthesized by SV3D, including side views that are not provided as inputs.
- We open-source the unofficial implementation of the training code for SV3D and preprocessing pipeline for Objaverse dataset, which are not officially released.

2. Related works

2.1. Novel View Synthesis

Novel View Synthesis(NVS) aims to generate new views of a scene or object from a limited view point. Diffusion models have emerged as a powerful method to generate the unseen viewpoints by leveraging its strong prior about the 3d understanding of the world. Works such as [5, 7, 8] have incorporated image and video diffusion models for NVS, and have obtained high quality novel views from the unseen viewpoints.

3. Method

To begin with, we implement the SV3D training code and fine-tune the model on our custom small dataset to verify that the code functions correctly (Sec. 3.1). Subsequently, we dive into NVR (Sec. 3.2) to refine the SV3D results, and discuss the specific architecture of the fusion layers (Sec. 3.3), which is a key component of our framework.

3.1. SV3D fine-tuning

We first implement the SV3D training code[1] and the data preprocessing pipeline[2] from scratch and open-source them, since there is no official implementation. To investigate the correctness of the implementation, we fine-tune the model on our own small dataset and check the results.

[1] https://github.com/jjihwan/SV3D-fine-tune

[2] https://github.com/briankwak810/objaverse_dataloader

3.2. Novel View Refinement

To refine the novel views synthesized by SV3D, we propose the Novel View Refinement (NVR) framework, leveraging the rich knowledge of the baseline model. Fig. 1 shows the conceptual illustration of NVR. Our refinement strategy can be decomposed as two stages. First, a front view of the object is fed into the original SV3D, and the novel views around the object are predicted (Fig. 1 top). However, the views near the back side might not be accurate since the front view lacks detailed information of the other views. Thus, in the second stage, a new back view of the object is fed into a second model (Fig. 1 botton), which is frozen and has same parameters as the original SV3D, to refine the inaccurate novel views near the back. For simplicity, we denote the first stage model as SV3D and the second stage as NVR. Meanwhile, the attention maps of SV3D, that is expected to have rich information of the front view, are injected into NVR after passing through trainable fusion layers (Sec. 3.3). Then, NVR can refine the results, provided the entire information of front view—from attention features—and the back view from the input image.

3.3. Fusion layers

For the attention features of original SV3D, A_{prev} and new ones of NVR, A_{curr}, fusion layers replace the attention maps of NVR as A_{mix}, to get rich information of SV3D. The precise definition of A_{mix} is

$$A_{\text{mix}} = B \odot \text{Conv}(P_\sigma A_{\text{prev}}) + (1 - B) \odot \text{Conv}(A_{\text{curr}}), \tag{1}$$

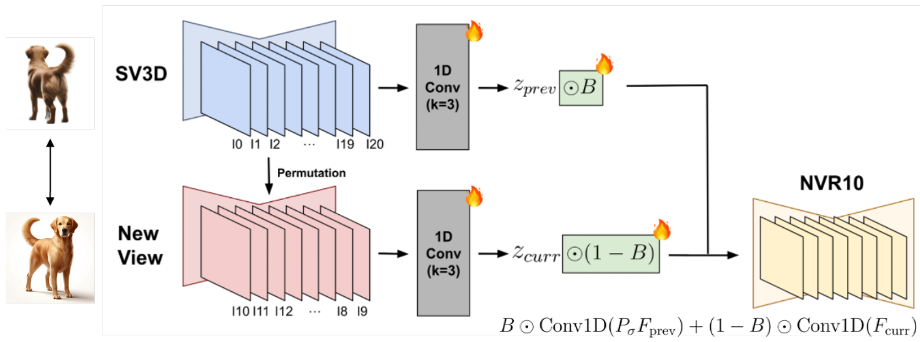

Figure 2. An illustration of Fusion Layers. Fusion layers involves permutation, temporal convolution, and blending layers. The attention maps of SV3D and NVR pass through the 1D convolution layers performed along the frame axis (i.e. temporal convolution) and mixed by blending layers. Before the temporal convolution, the attention maps of SV3D are multiplied by permutation matrix, P_σ, to align the frame indices with NVR.

where P_σ, Conv, and B denote permutation, temporal convolution, and blending layers, respectively.

Permutation layers The Permutation layer reorders the attention features of SV3D to align with NVR that receive back view as an input. Since we utilize 10^{th} frame as input of NVR model, the corresponding permutation matrix is defined as follows:

$$P_\sigma = \begin{bmatrix} 0_{10\times 11} & I_{10} \\ I_{11} & 0_{11\times 10} \end{bmatrix}. \tag{2}$$

Note that the permutation matrix is not a trainable parameter, to strictly align the attention features.

Temporal convolution layers After aligning the temporal indices of the two attention maps—SV3D and NVR—, they pass through a 1D convolution along the frame axis with the kernel size of 3, to properly refer to the attention maps of adjacent frames. Specifically, each kernel of the channel are explicitly initialized as $[0, 1, 0]$, to force them to only focus on themselves at the beginning, and gradually referring to information of adjacent frames during training.

Blending layers After passing through the temporal convolution, they are blended by the blending parameters, $B \in \mathbb{R}^{21}$. The two processed attention maps are alpha-blended frame-wisely. Especially, since the attention maps from SV3D contain much more important information of the front view, the blending parameters are initialized as follows:

$$B = [0, \frac{1}{11}, \cdots, \frac{10}{11}, 1, \frac{10}{11}, \cdots, \frac{1}{11}, 0]. \tag{3}$$

4. Experiments

In this section, we provide implementation details and qualitative results that demonstrate correctness and effectiveness of our implementation and proposed framework.

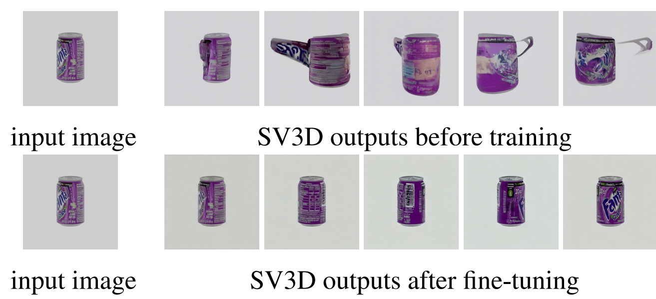

Figure 3. Qualitative results before (top) and after (bottom) SV3D fine-tuning.

4.1. Implementation details

Dataset We use the widely used Objaverse dataset[2]. Since the Objaverse datset is saved as a mesh format, we preprocess the data to have arbitrary elevation and center of mass.

Implementation Details First, we implement the training code for the SV3D model based on the implementation details in the Stable Video Diffusion paper[1]. Pytorch Lightning and Deepspeed was used to speed up and save the computational cost. We trained the SV3D model with 73 pre-processed datapoints from the Objaverse dataset, with 1 A6000 GPU for 100k steps with a batch size of 1. We use the AdamW optimizer [6] with a fixed learning rate of $1e^{-5}$.

For the NVR model training, first we run inference using the SV3D model and obtain the spatial attention features (both the self and cross attention) of the intermediate Unet layers for each timestep.

Next we select the back view image from the ground truth dataset, which is used as the first frame condition for our NVR pipeline. Also the ground truth video latents are rolled to match the indices with the back view image. We

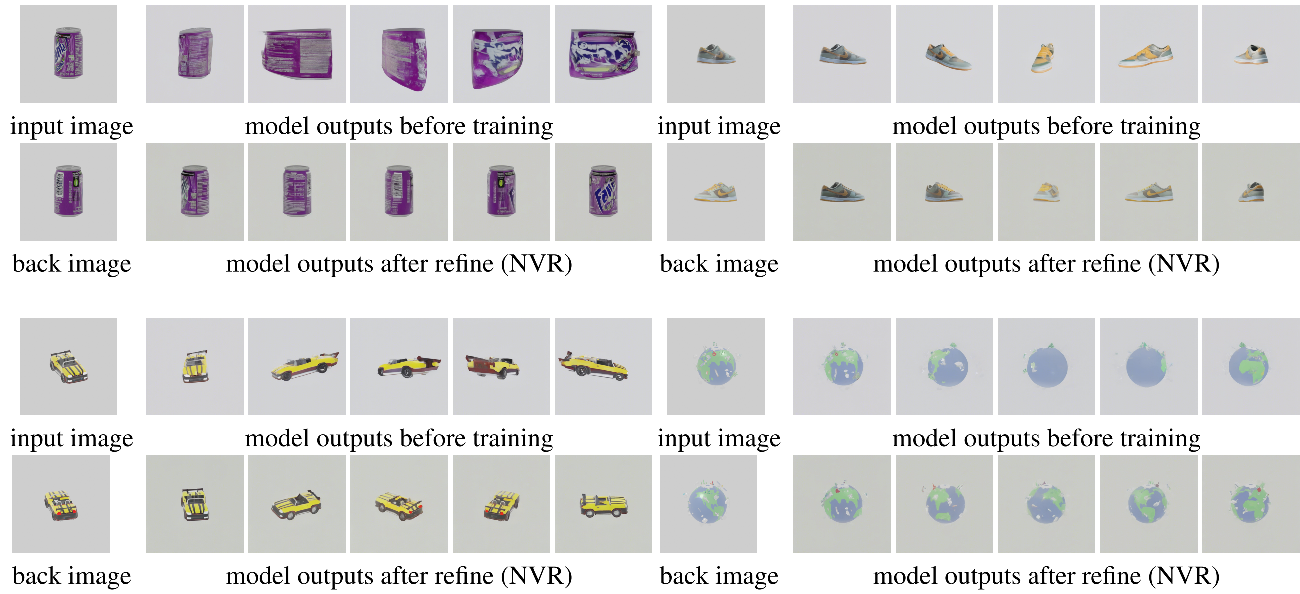

Figure 4. Qualitative results before (top row) and after (bottom row) NVR traininig. The results show that our NVR framework effectively refine the original novel views using a new back view image.

follow the widely-used EDM framework[4] in our training process, which involves continuous time forward process. Since the inference of the baseline model only provides the attention features of discrete timesteps, we first sample the continous timestep and pick the corresponding discrete timestep with the minimum L1 loss between them. The corresponding attention features go through our proposed trainable fusion layers and are injected into our NVR pipeline. All of the weights of the NVR model is freezed during the training, and only the fusion layers are the trainable parameters, resulting in a big reduction in the number of parameters as shown in Tab. 1. We train the model with 5 datapoints from the objavarse dataset, with 1 A6000 GPU for 10k steps with a batch size of 1. We use the AdamW optimizer with a fixed learning rate of $1e^{-3}$.

model	SV3D	NVR
parameters	1.5B	150k

Table 1. Model parameters comparison

4.2. Results

Finetuning the SV3D model showed striking quality improvements as shown in Fig. 3. Also fine tuning our NVR model shown improvements in both quality consistency especially for the unseen viewpointsFig. 4, as similar as finetuning the whole model. Therefore we have shown that our model can improve the performance the novel views by only using 150k parameters.

5. Conclusion

We have proposed a novel framework called Novel View Refinement (NVR) that provides a simple yet effective way to boost novel view synthesis performance while being parameter efficient. Our proposed fusion layer architecture, which efficiently combines the attention maps from the two stages works requiring only small numbers of trainable paramters.

Our qualitative experiments on the Objaverse dataset demonstrate that NVR can effectively refine the synthesized novel views from SV3D, improving quality not just of the back views but also the side views. This showcases the power of leveraging the learned knowledge in the pretrained model attention maps.

We hope this work inspires further research into refining large vision synthesis models in a computationally feasible manner, and look forward to real-world applications that use novel view refinement in a more generalizable way.

6. Limitations

Due to the lack of computational resources, such as VRAM or NFS memory, we could not scale up to large datasets. We plan to scale up our model to the entire Objaverse XL dataset to realize generalizable inference on unseen data.

We also plan to experiment multiple design choices to find optimal architecture of fusion layers with higher generalizability and less training parameters.

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